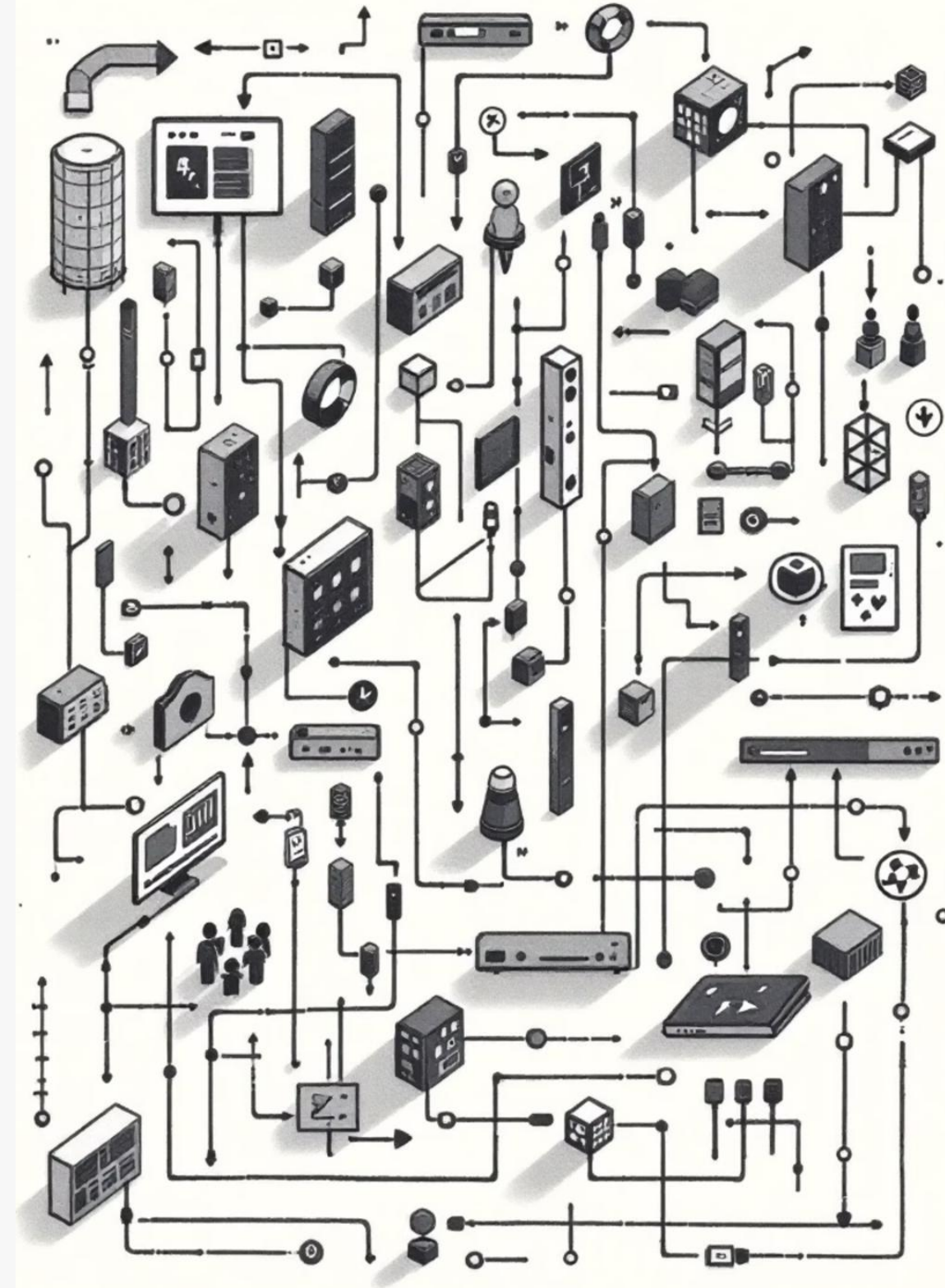




How Service Registry Works (SOAP & REST Context) and Tools for JSON-based Service Registry

Group 11

CCS2552 – Service Oriented Web Applications



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The Chaos of Hardcoded Connections

In a microservices world, directly linking services creates a brittle and unmanageable system.



Static IP Addresses

Hardcoding IP addresses or port numbers leads to rigid dependencies.



Service Failures

When a service instance fails or moves, all dependent services break.



Scaling Challenges

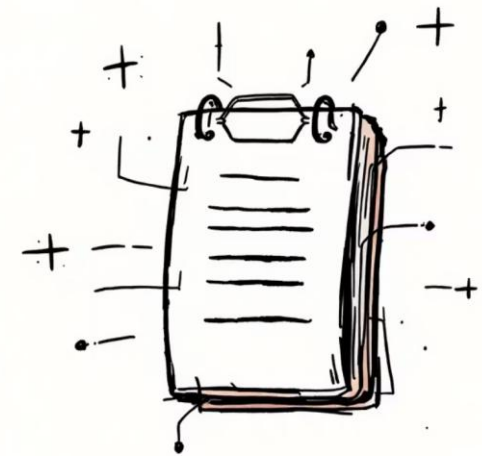
Adding or removing service instances requires manual updates across the entire system.

What is a Service Registry?

Think of a service registry as the dynamic “Phonebook”, in modern terms a “Data Base” for your microservices, enabling seamless discovery and communication.

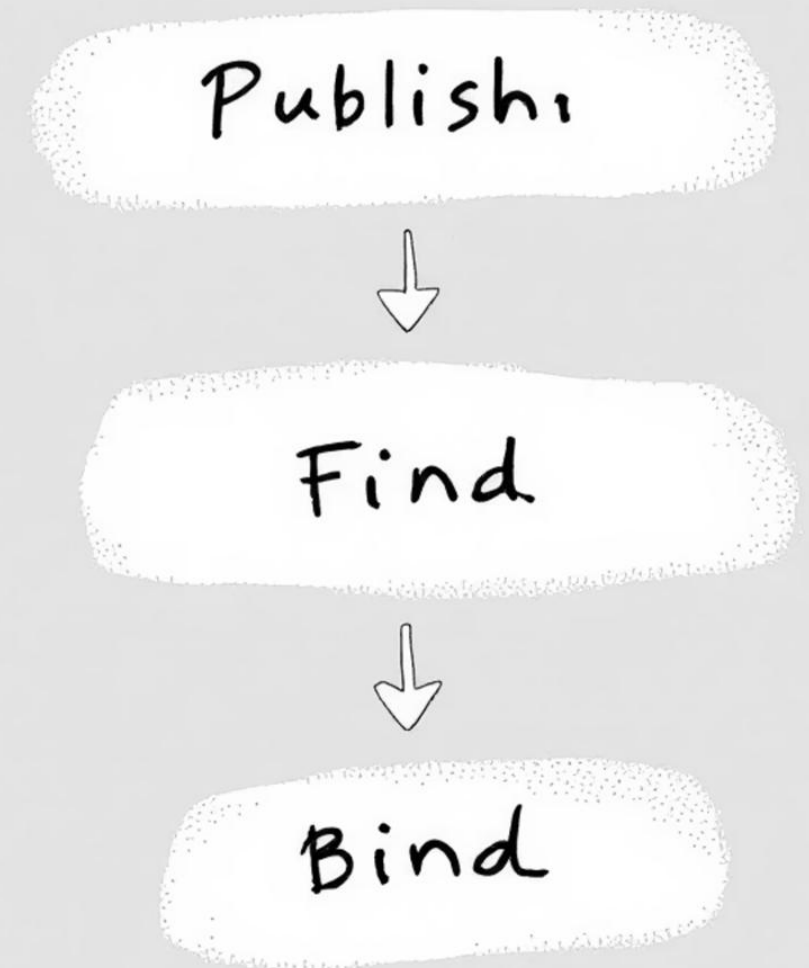
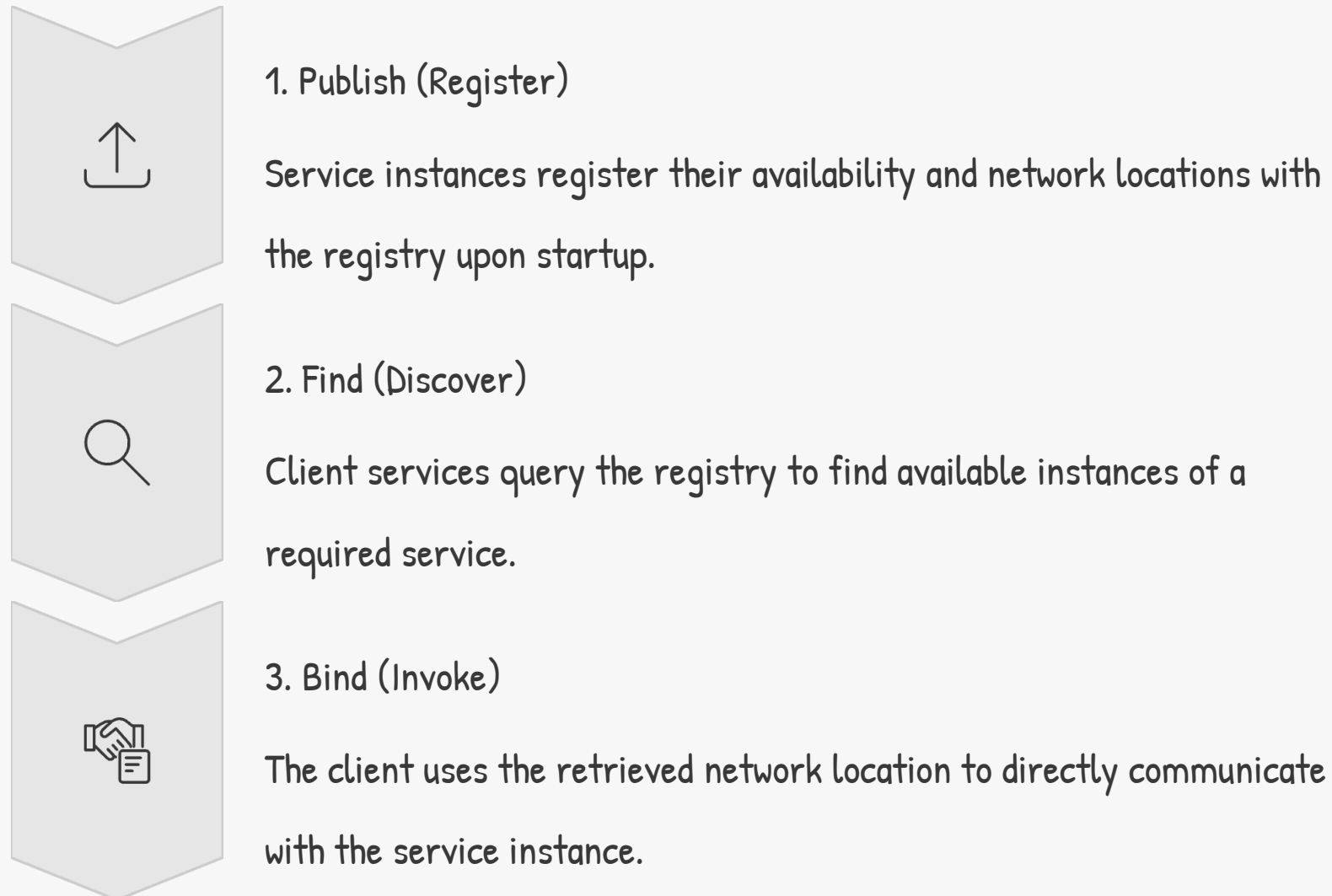
A centralized database that stores the network locations of all service instances.

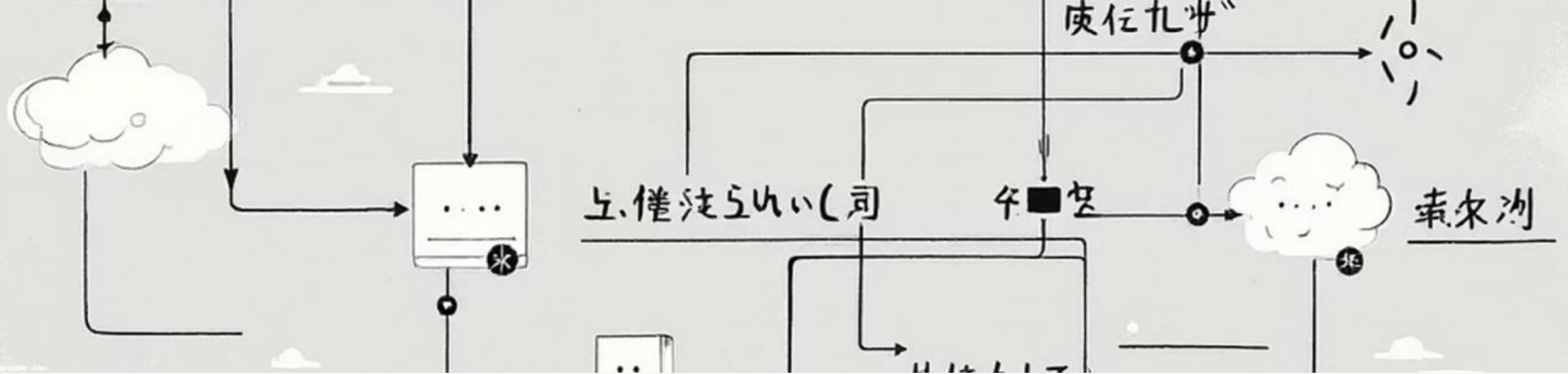
- Acts as a single source of truth for service availability.
- Crucial for dynamic, scalable, and resilient microservices architectures.
- Decouples service providers from service consumers.



The Service Registry Mechanism

A three-step process ensures services can register themselves, be found, and invoked by others.





SOAP: The Protocol-Centric Era

Historically, SOAP-based services relied on a protocol-centric approach with strict contracts for discovery.

UDDI (Universal Description, Discovery, and Integration)

A registry for publishing and discovering web services, primarily SOAP-based.

WSDL (Web Services Description Language)

XML-based format describing the functionality, input/output, and location of a web service.

Strict Contracts

Services and clients were bound by predefined XML schemas, ensuring rigid interoperability.

SOAP vs. REST: A Comparative View

Understanding the fundamental differences in their approach to service interaction and discovery.

SOAP (Simple Object Access Protocol)

- Contract-focused (WSDL, XML Schema)
- Message-based (XML envelopes)
- Strongly typed
- Stateful or stateless
- Tightly coupled

REST (Representational State Transfer)

- Location-focused (URLs)
- Resource-based (JSON, XML, plain text)
- Loosely typed
- Stateless (preferred)
- Loosely coupled

Why JSON for Modern Registries?

JSON's inherent qualities make it the ideal format for contemporary service registries.

- ❖ **Human-Readable:** Easy for developers to read and write, simplifying debugging.
- ❖ **Web-Native:** Natively supported by web browsers and APIs, enabling seamless integration.
- ❖ **Lightweight & Efficient:** Smaller payload size compared to XML, leading to faster data transfer.
- ❖ **Flexible Schemas:** Adapts well to evolving service metadata without rigid schema enforcement.



A hand-drawn illustration on a light blue background. At the top, the word "JSON" is written in large, bold, black capital letters. Below it, a JSON object is written in a smaller, black, monospace font: "{ \"name\": \"John\", \"age\": 30, \"city\": \"New York\" }". At the bottom, the text "Simptok coscens!" is written in a small, black, cursive font.

Modern JSON-Based Service Registry Tools

Key tools empowering dynamic service discovery in today's microservices landscape.



1) Consul

HashiCorp's solution offering service discovery, configuration, and segmentation functionality.



2) Eureka

Netflix's highly available service registry and discovery server, part of the Spring Cloud ecosystem.



3) etcd

A distributed reliable key-value store, commonly used for shared configuration and service discovery.

Importance of investigation

Studying registry types uncovers the hidden backbone that enables modern systems to scale and operate without failures.



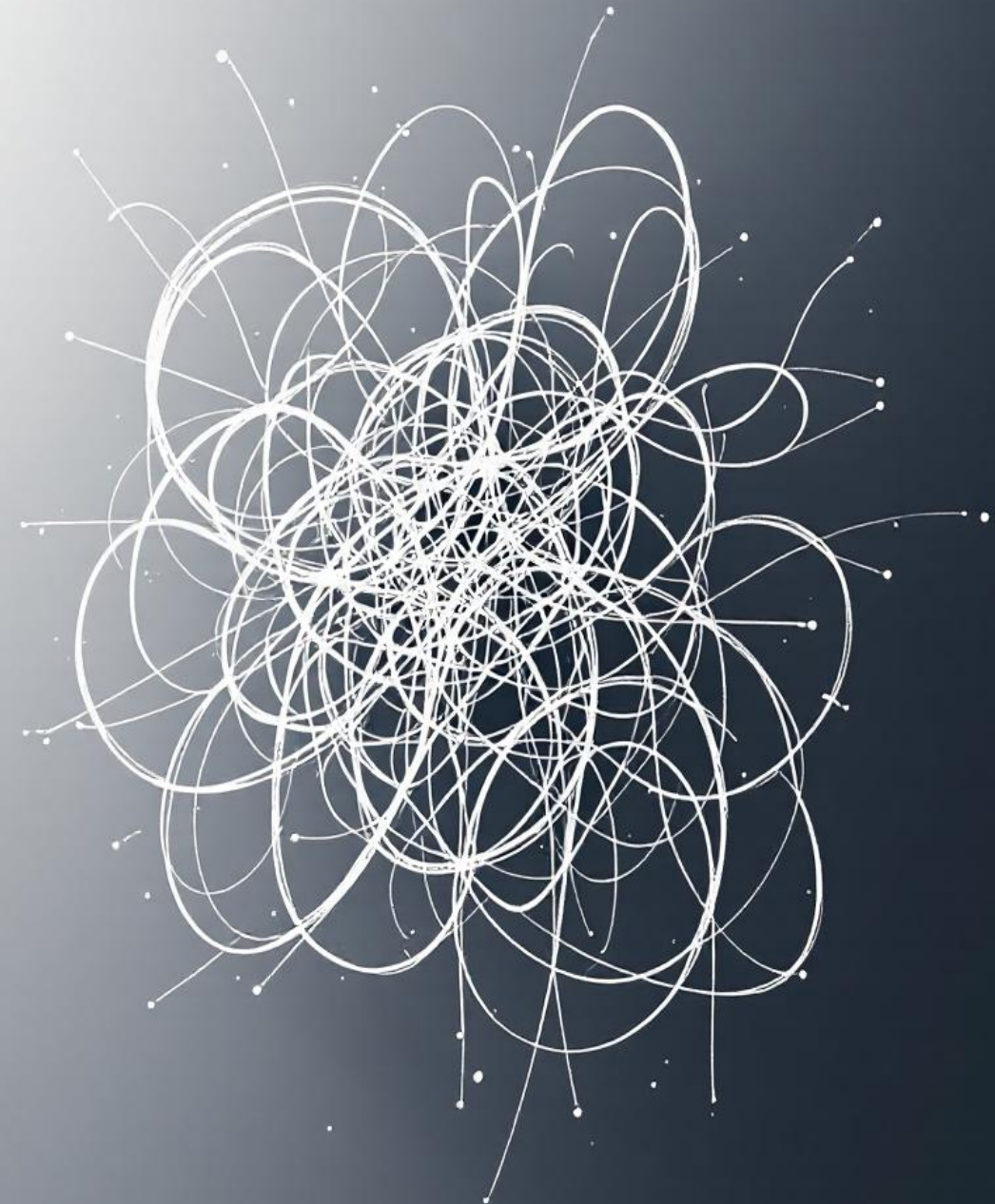
Service Registry Types Guide Modern System Architecture

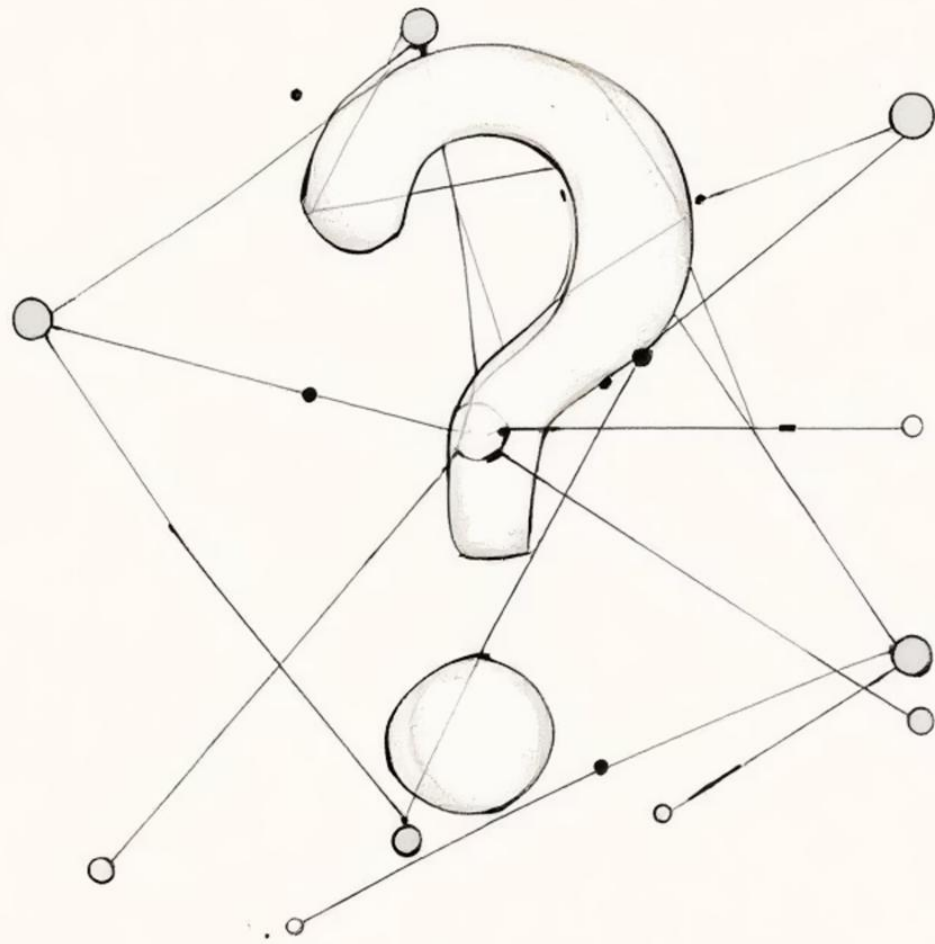
Dynamic, centralized registries prevent failures and simplify communication across distributed services.



JSON Registries Power Modern Cloud-Native Discovery

Tools like Consul, Eureka, and etcd deliver lightweight, scalable, and highly dynamic service discovery.





Questions & Discussion

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Thank
You!

